Methodology Passport Card

1. Project Information

- **Title:** Main factors affecting football players' market value in Big-5 European leagues: an econometric analysis by position groups
- **Authors:** Mukhamedzhan Kadir, Yerkanat Manapuly
- **GitHub Repository Link:** https://github.com/muhamed434/big5-player-valuation-econometrics

2. Type of Research

- **Type:** Quantitative Research (Econometric Analysis).
- **Justification:** The study relies on secondary quantitative statistical data collected from Transfermarkt and Fbref/StatsBomb. Econometric modeling (Pooled OLS, Fixed Effects, polynomial age terms) is applied to empirically test formal hypotheses regarding market value determinants across player positions.

3. Hypotheses Formulation

Research Question 1 (RQ1: Positional Valuation Differences)

- **Null Hypothesis (H0):** Controlling for playing time, league, and contract terms, the average log market value of players does not differ significantly across position groups (defenders, midfielders, forwards).
- **Alternative Hypothesis (H1):** Holding playing time, league, club status, and remaining contract length constant, forwards have a statistically significantly higher average log market value compared to defenders and midfielders.

Research Question 2 (RQ2: Non-linear Age Impact and Positional Peaks)

- **Null Hypothesis (H0):** Age has a strictly linear effect on market value, and the age of reaching peak market value is identical across all position groups.
- **Alternative Hypothesis (H1.1 - Non-linearity):** The effect of age on market value is non-linear (parabolic), where market value increases with age up to a specific peak, after which it declines.
- **Alternative Hypothesis (H1.2 - Positional Peak Differences):** The peak market value for defenders is reached at an older age (26 to 28 years old) than for attacking players (24 to 26 years old).

Research Question 3 (RQ3: Position-Specific Performance Metrics)

- **H0 (Null Hypothesis):** Position-specific defensive metrics do not yield a higher explanatory power (Adjusted R^2) for defenders' market value compared to standard aggregate "goals plus assists" metrics (i.e., Adj R^2_defensive $\leq$ Adj R^2_standard).
- **H1 (Alternative Hypothesis):** Position-specific defensive metrics yield a statistically significant higher explanatory power (Adjusted R^2) for defenders' market value than standard aggregate "goals plus assists" metrics (i.e., Adj R^2_defensive > Adj R^2_standard).

4. Matrix of Variables

Category	Variable Name	Description & Measurement Unit
Dependent Variable	log_market_value	Natural logarithm of player's market value in EUR: ln(Market Value)
Independent Variables	position	Categorical position group (Defender, Midfielder, Forward)
	age, age^2	Player age in years and quadratic age (Age squared) to model non-linear trajectories
	position_x_age	Interaction terms between age and position group
	tackles_per90	Successful tackles per 90 minutes (Defenders)
	interceptions_per90	Interceptions per 90 minutes (Defenders)
	duel_win_pct	Ground and aerial duel win percentage (%) (Defenders)
	prog_passes_per90	Progressive passes per 90 minutes (Midfielders)
	xg_per90	Non-penalty Expected Goals (xG) per 90 minutes (Forwards)
Control Variables	league	League fixed effects (EPL, La Liga, Serie A, Bundesliga, Ligue 1)

	minutes_played	Total official playing minutes in the season
	club_tier	Club financial status / league table finish tier
	contract_years	Remaining contract length in years

5. Selected Metrics and Baseline

- **Primary Metric:** Adjusted R-Squared ($\text{Adj } R^2$) — Used as the primary measure of model fit and explanatory power when comparing nested and position-specific regression specifications.

- **Secondary Metric:** Root Mean Squared Error (RMSE) — Measures the average prediction residual of market value estimation in log units.

- **Statistical Decision Criterion:** p-value threshold ($p < 0.05$) — Two-tailed t-tests for individual coefficient significance and F-tests for overall model explanatory gains.

- **Guardrail Metrics:**

    1. Multicollinearity: Variance Inflation Factor ($\text{VIF} < 5.0$).
    2. Heteroskedasticity: Breusch-Pagan test p-value > 0.05.
    3. Sample Size Constraint: Minimum $N \ge 300$ observations per position group.

- **Baseline Model:** A standard baseline Ordinary Least Squares (OLS) model estimating player log market values using aggregate performance metrics (`goals_assists_per90`) without position-specific sub-group interactions.